

Introduction to Statistical Investigations

Second Edition

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Chapter 3.1

Estimation

Statistical Inference: Confidence Intervals

Chapter 3 Overview

- So far, we can only say things like
 - “We have strong evidence that the probability Harley chooses the correct cup is > 0.5 .”
 - “Students overestimate the length of the Jackson 5’s song ‘ABC.’”
- We want a method that estimates what the parameter *actually is*.

Can Dogs Sniff Out Cancer?

Marine, a dog originally trained for water rescues, was tested to see if she could detect if a patient had colorectal cancer by smelling a sample of the patient's breath.

Can Dogs Sniff Out Cancer?

- In each trial, she first smells a bag from a patient with colorectal cancer.
- Then she smells 5 other samples; 4 from normal patients and 1 from a person with colorectal cancer.
- Marine correctly identified the bag from the cancer patient in 30 out of 33 trials. So our sample proportion is $30/33 = 0.909$.

Can Dogs Sniff Out Cancer?

- In each of the 33 trials, Marine smells 5 bags containing breath samples: 4 from normal patients and 1 from a person with colorectal cancer.
- Discuss:
 - What are the observational units?
 - What is the variable?
 - What is our sample size?
 - What's the general probability of success, π ?
 - Write the null and alternative hypotheses in symbols.

Can Dogs Sniff Out Cancer?

Our sample proportion lies more than 10 standard deviations above the mean and hence our p-value is 0.

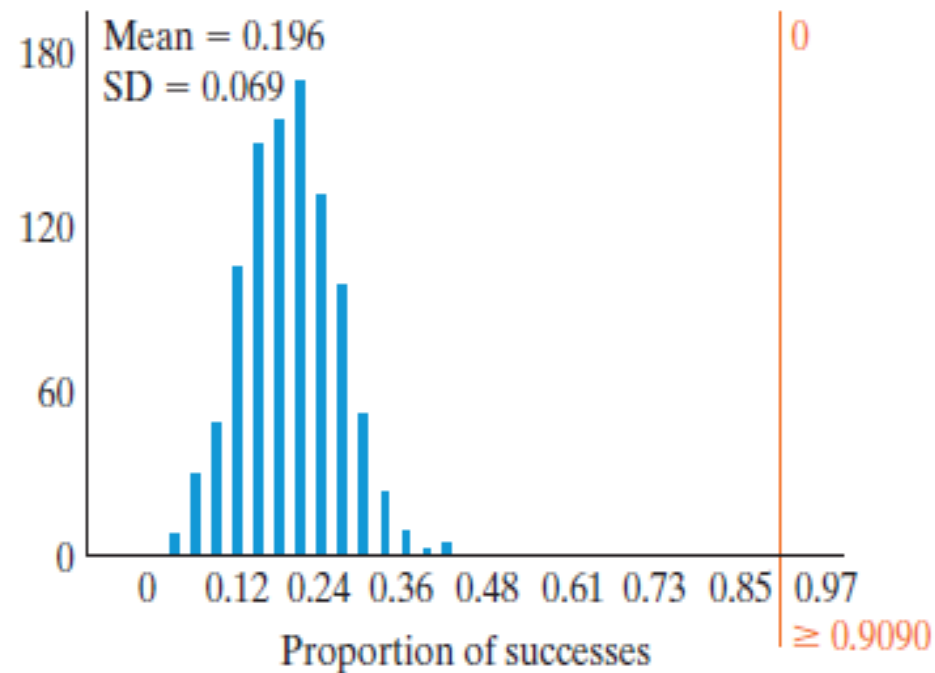
Probability of success (π):

Sample size (n):

Number of samples:

As extreme as

Proportion of samples:
0/1000 = 0



Can Dogs Sniff Out Cancer?

- It's not very interesting to say “Marine picks the correct bag more than 20% of the time.”
- Can we say for certain that Marine picks the correct bag 90.9% of the time?
- Is it plausible that Marine actually picks the correct bag 70% of the time and just had a lucky day?

Can Dogs Sniff Out Cancer?

- Let's try:
- $H_0: \pi = 0.70$ $H_a: \pi \neq 0.70$
- We get a small p-value (0.0090) so we can rule out 0.70 as a value for π .

Probability of success (π):

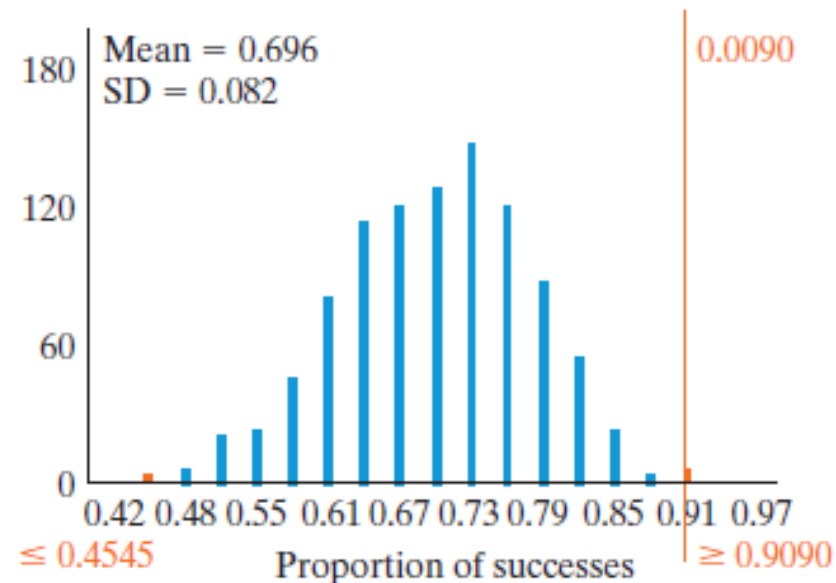
Sample size (n):

Number of samples:

As extreme as

Proportion of samples:
(3 + 6)/1000 = 0.0090

☒ Two-sided



Can Dogs Sniff Out Cancer?

- What about 0.80?
- Is an observed statistic of 0.909 unlikely if $\pi = 0.80$?

Can Dogs Sniff Out Cancer?

- $H_0: \pi = 0.80$ $H_a: \pi \neq 0.80$
- We get a large p-value (0.1470) so 0.80 is a ***plausible*** value for Marine's long-run probability.

Probability of success (π):

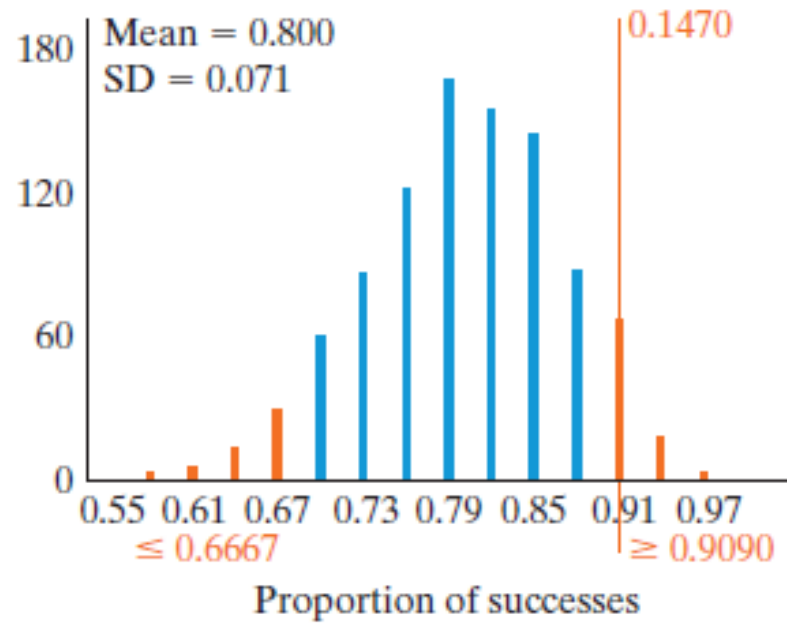
Sample size (n):

Number of samples:

As extreme as

Proportion of samples:
(52 + 95)/1000 = 0.1470

☒ Two-sided



A range of plausible values

- Small p-value $\Rightarrow$ the value in H_0 is not plausible. (We can conclude H_a .)
- Large p-value $\Rightarrow$ the value under H_0 is plausible. (We can't conclude H_a .)
- The **significance level** α determines whether a p-value is small or large.

A range of possible values

- We can use the One Proportion applet to find a range of plausible values for Marine's long-term probability of choosing the correct bag.
- We do this by keeping the sample proportion the same and varying π .
- You will do this in today's Exploration.

Exploration 3.1

3.2 and 3.3

Overall Goal

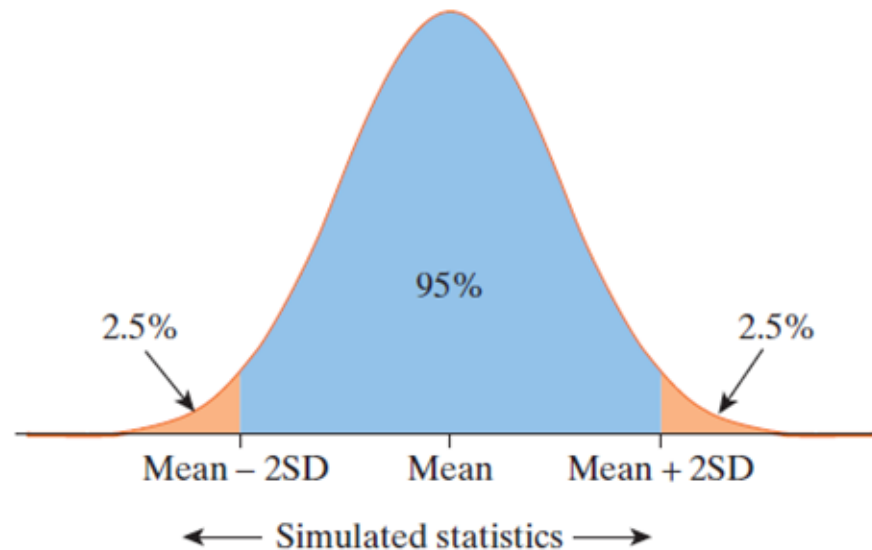
- Estimate the *population proportion* or *population mean*.
- How to do this:
 - Collect a sample
 - Obtain a sample proportion or sample mean (depending on what we are measuring)
 - Determine where the population proportion or mean lies with a certain level of confidence.
- It's tedious to construct the confidence interval by trial and error.

Solution: Theory

- We will use a theory-based method to accomplish this.

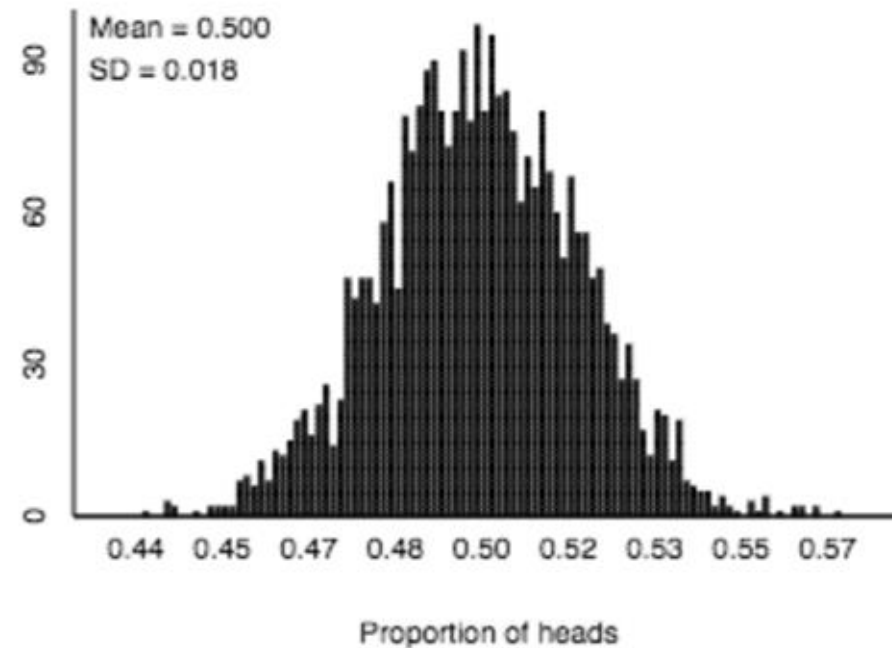
Example 3.2: 2SD Method (1 of 4)

- When a null distribution is bell-shaped, about 95% of the statistics will fall within 2 standard deviations of the mean with the other 5% outside this region



Example 3.2: Where Do We Get the SD?

- A null distribution for a poll about cyberbullying with $n = 743$ and $\pi = 0.50$ gives a SD of 0.018.



Example 3.2: 2SD Method (2 of 4)

- So we could say that a parameter value is plausible if it is within 2 standard deviations (SD) from our best estimate of the parameter, our observed sample statistic.
- This gives us the simple formula for a 95% confidence interval of

$$\hat{p} \pm 2SD$$

Example 3.2: 2SD Method (3 of 4)

- If our observed statistic is 0.59,
- Using the 2SD method to create an approximate 95% confidence interval we get:

$$0.59 \pm 2(0.018)$$

$$0.59 \pm 0.036$$

- The 0.036 in the above is called a **margin of error**.
- The interval can also be written as we did before using just the endpoints; (0.554, 0.626)
- This is approximately what we got with our range of plausible values method (a bit wider).

Example 3.2: 2SD Method (4 of 4)

- We used $\pi = 0.50$ and $n = 743$ to develop the null distribution we used to get the standard deviation needed for our 2SD method.
- While the sample size of 743 came from the survey, the parameter of 0.50 did not.
- We used 0.50 because using that gives us the largest possible standard deviation and largest possible margin of error.
- This means we are at least 95% confident that our interval contains the population parameter.

Example 3.2: Theory-Based Methods (1 of 2)

- The 2SD method only gives us an approximate 95% confidence interval
- If we want a different level of confidence, we can use the range of plausible values (hard) or theory-based methods (easy).
- In Chapter 1, we saw that the SD of $\hat{p}$ was:

$$\sqrt{\frac{\pi(1-\pi)}{n}}$$

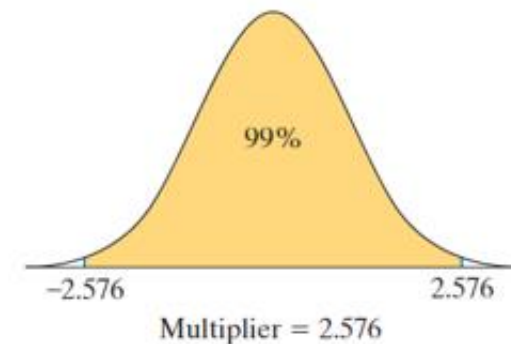
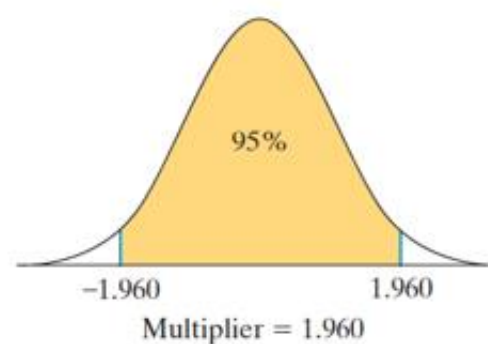
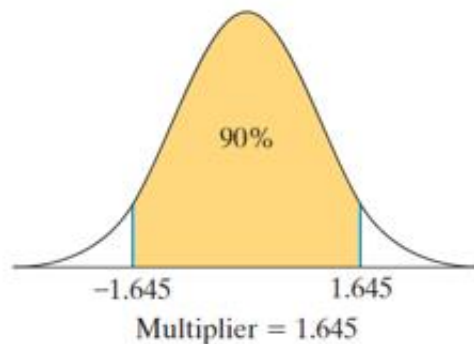
Example 3.2: Theory-Based Methods (2 of 2)

- We don't know the value of the parameter, π , so we substitute $\hat{p}$ in its place.
- We also might want to adjust the confidence level, so we will not always use 2 (or approximately 2) for our multiplier.
- Therefore we can develop a formula for confidence intervals:

$$\hat{p} \pm \text{multiplier} \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

Example 3.2: How is the multiplier found?

- The multiplier comes from the standard normal distribution (with mean 0 and standard deviation 1).
- Below shows corresponding multipliers for different confidence levels.



Example 3.2: Two ways to write a Confidence Interval

- Suppose you are told a 95% confidence interval for some population proportion is 0.542 to 0.634. How do you rewrite it in terms of center $\pm$ margin of error?
- The center (which should be the statistic) is halfway between the endpoints: $\frac{0.542 + 0.634}{2} = 0.588$.
- The margin of error is distance from an endpoint to the center: $0.634 - 0.588 = 0.046$.
- We rewrite the interval as 0.588 ± 0.046 .

What about quantitative variables?

Example 3.3: Confidence Intervals (1 of 2)

- Essentially, replace $\hat{p}$ with $\bar{x}$
- Remember the basic form of a confidence interval is:
statistic $\pm$ multiplier $\times$ (SD of statistic)
- In our case, the statistic is $\bar{x}$ so we write our 2SD confidence interval as:

$$\bar{x} \pm 2(\text{SD of } \bar{x})$$

Example 3.3: Confidence Interval (1 of 3)

- The confidence interval for a population mean has the form:

$$\bar{x} \pm \text{multiplier} \times \frac{s}{\sqrt{n}}$$

- Where
 - $\bar{x}$ is the center (our best estimate for the population mean)
 - $\frac{s}{\sqrt{n}}$ is the predicted value of the standard deviation of the null distribution (also called the standard error of the statistic)
 - Multiplier is a number that is dependent upon the confidence level (like ≈ 2 for 95% confidence intervals)